prober–retaliator is basically like a retaliator, but he occasionally tries a brief experimental escalation of the contest. He persists in this hawk-like behaviour if his opponent does not fight back. If, on the other hand, his opponent does fight back he reverts to conventional threatening like a dove. If he is attacked, he retaliates just like an ordinary retaliator.

If all the five strategies I have mentioned are turned loose upon one another in a computer simulation, only one of them, retaliator, emerges as evolutionarily stable.* Prober–retaliator is nearly stable. Dove is not stable, because a population of doves would be invaded by hawks and bullies. Hawk is not stable, because a population of hawks would be invaded by doves and bullies. Bully is not stable, because a population of bullies would be invaded by hawks. In a population of retaliators, no other strategy would invade, since there is no other strategy that does better than retaliator itself. However, dove does equally well in a population of retaliators. This means that, other things being equal, the numbers of doves could slowly drift upwards. Now if the numbers of doves drifted up to any significant extent, prober–retaliators (and, incidentally, hawks and bullies) would start to have an advantage, since they do better against doves than retaliators do. Prober–retaliator itself, unlike hawk and bully, is almost an ESS, in the sense that, in a population of prober–retaliators, only one other strategy, retaliator, does better, and then only slightly. We might expect, therefore, that a mixture of retaliators and prober–retaliators would tend to predominate, with perhaps even a gentle oscillation between the two, in association with an oscillation in the size of a small dove minority. Once again, we don't have to think in terms of a polymorphism in which every individual always plays one strategy or another. Each individual could play a complex mixture between retaliator, prober–retaliator, and dove.

This theoretical conclusion is not far from what actually happens in most wild animals. We have in a sense explained the 'gloved fist' aspect of animal aggression. Of course the details depend on the exact numbers of 'points' awarded for winning, being injured, wasting time, and so on. In elephant seals the prize for winning may be near-monopoly rights over a large harem of females. The pay-off for winning must therefore be rated as very high. Small wonder that fights are vicious and the probability of serious injury is also high. The cost of wasting time should presumably be regarded as small in comparison with the cost of being injured and the benefit of winning. For a small bird in a cold climate, on